

R.A.I. - COMPLETE TECHNICAL KNOWLEDGE BASE & VIVA MASTER MANUAL

Smart India Hackathon (SIH) Problem Statement: SIH1521 - Explainable AI for Heavy/High-Impact Rainfall Prediction

Version: 1.0.0-PROD-FROZEN | Baseline Commit: b70c17a | Team Distribution: 6 Members

Problem Statement: SIH1521 - Explainable AI for Heavy/High-Impact Rainfall Prediction

Target Audience: All 6 Team Members (Easy to Read, Understand & Memorize)

Tone: Simple, Clear, Step-by-Step, Real-World Analogies

Version: 1.0.0 (Pre-Deployment Freeze Baseline)

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CHAPTER 1: WHAT IS R.A.I. & WHY DOES IT EXIST?

1. What is R.A.I.?

R.A.I. stands for **Rainfall Artificial Intelligence**.

It is an intelligent weather website and early-warning platform that predicts **heavy rainfall 24 hours in advance** and explains **WHY** it is going to rain using Explainable AI (XAI).

2. What problem are we solving?

- In India, sudden heavy rains and cloudbursts cause flash floods, drown city streets, and destroy farmers' crops.
- Regular weather apps (like Google Weather or Apple Weather) give vague forecasts like "40% chance of rain" or "Light rain", but they **do not predict extreme flood-level storms** accurately and **never explain why**.
- Normal AI models are "black boxes" - they give a number, but nobody knows if they can be trusted.

3. What is the SIH Problem Statement?

- **Problem Statement ID:** `SIH1521`
- **Title:** **"Explainable AI for Heavy/High-Impact Rainfall Prediction"**
- **The Goal:** Create an AI model that predicts heavy rain (≥ 64.5 mm/day, as defined by the IMD) and gives human-readable explanations using mathematical XAI (TreeSHAP).

4. Who uses R.A.I.?

1. **Regular Citizens:** To know if their city or neighborhood is at risk of waterlogging or heavy storms.
2. **Farmers:** To get customized farming advice in Hindi/English based on their specific crop and soil (e.g. "Don't spray fertilizer today because heavy rain will wash it away").
3. **Disaster Management Teams (NDRF/SDRF):** To see live radar maps, flood risk gauges, and deploy rescue boats/pumps before the flood hits.

CHAPTER 2: HOW THE WHOLE SYSTEM WORKS (SIMPLE 4-STEP STORY)

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STEP 1: FETCH DATA
User opens R.A.I. and selects a city (e.g., Kanpur).
Our backend fetches real-time atmospheric data from Open-Meteo & NASA GPM satellites.
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STEP 2: PREDICT RISK WITH ML
Our XGBoost Machine Learning model analyzes 14 atmospheric features
(temperature, pressure, humidity, dew point, wind, etc.) and predicts rain probability.
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STEP 3: EXPLAIN WITH TreeSHAP (XAI)
TreeSHAP calculates exact mathematical contributions for every feature:
- "High humidity (+0.41) increases flood risk."
- "Stable air pressure (-0.28) protects against cloudbursts."
|
v
STEP 4: SHOW ACTIONABLE WARNINGS
The frontend shows a 4-Layer Inspector:
[Layer 1: Observed Weather] -> [Layer 2: Predicted Probability] ->
[Layer 3: TreeSHAP Explanations] -> [Layer 4: Action Checklist for Citizens & Farmers]

```

CHAPTER 3: THE COMPLETE TECH STACK (PLAIN ENGLISH)

Frontend (What you see in the browser)

- **React 18:** The JavaScript library used to build the user interface with interactive components.
- **TypeScript:** Adds strict type checking to JavaScript to prevent bugs.
- **Vite:** A fast bundler and local development server (`http://localhost:3000`).
- **CSS Modules:** Keeps styling clean and scoped to individual components without style conflicts.
- **Leaflet & RainViewer:** Displays the interactive map with real-time Doppler precipitation radar animations.

Backend (The engine that runs behind the scenes)

- **Python (3.11 / 3.14):** The core programming language used for machine learning and APIs.
- **FastAPI:** A high-speed web framework that receives requests from the frontend and returns predictions. Runs on port `8000`.
- **Uvicorn:** The lightning-fast ASGI server that runs our FastAPI code.
- **Pydantic:** Validates all incoming and outgoing data to make sure coordinates and numbers are correct.

Machine Learning & Explainable AI

- **XGBoost**: The Gradient Boosted Decision Tree algorithm that makes the heavy rainfall predictions.
- **Scikit-Learn**: Used for probability calibration (Platt scaling / Isotonic) and metric calculations.
- **SHAP (TreeSHAP)**: The Explainable AI library that calculates exact mathematical Shapley values.
- **NumPy & Pandas**: Used for tabular data manipulation and calculating rolling meteorological features.

Database & Authentication

- **Supabase Cloud PostgreSQL**: Managed cloud SQL database that stores users, farmer profiles, farm plots, and login sessions.
- **Brevo (Sendinblue)**: Transactional email service that sends real 6-digit OTP verification codes to user inboxes using verified Sender ID `1` (RAINAIWORK@GMAIL.COM).
- **Salted HMAC & SHA-256**: Cryptographic security ensuring passwords and OTPs are never stored in plaintext.

DevOps & Deployment

- **Docker**: Packages the FastAPI backend and ML models into a clean, lightweight Linux container (`python:3.11-slim`).
- **Git & GitHub**: Version control system used to track changes and preserve the frozen `v1.0` baseline.

CHAPTER 4: PROJECT STRUCTURE (WHICH FILE DOES WHAT?)

Here is the clean folder layout of our project:

Backend & Machine Learning (`ml/`)

- `ml/server.py`: The main FastAPI server file. Defines all API routes (`/api/prediction/\*`, `/api/auth/\*`, `/api/farmer/\*`).
- `ml/weather\_service.py`: Fetches real-time numerical weather data from Open-Meteo and NASA GPM satellite feeds.
- `ml/operational\_engine.py`: Translates raw probabilities into warning levels (`NO\_WARNING`, `ADVISORY`, `WATCH`, `WARNING`).
- `ml/inference/service.py`: Assembles the 14-feature vector and runs the XGBoost model.
- `ml/explainability/shap\_engine.py`: Runs TreeSHAP to calculate positive risk drivers and protective factors.
- `ml/auth/otp\_engine.py`: Generates 6-digit OTPs, creates salted HMAC hashes, and sends emails via Brevo.
- `ml/db/client.py`: Connects to Supabase Cloud PostgreSQL to save users, sessions, and farmer plots.
- `ml/models/rainfall\_model\_v1/`:
 - **model.pkl**: The trained XGBoost model binary (573 KB).
 - **calibrated_model.pkl**: The probability calibrator (Platt/Isotonic).
 - **metadata.json**: Model version, training date, and threshold info.
 - **metrics.json**: Test metrics (ROC-AUC 0.9605, PR-AUC 0.1082, Brier score 0.0044).

Frontend User Interface (`src/`)

- `src/main.tsx`: React application entry point.
- `src/App.tsx`: Sets up global providers, modals, and route containers.
- `src/routes/AppRoutes.tsx`: Defines all web page URLs (`/dashboard`, `/risk-map`, `/farmer`, `/emergency`, `/relief`).
- `src/pages/RiskMap/RiskMapPage.tsx`: **The Core SIH Showcase Page** with the interactive radar map and 4-layer telemetry inspector.
- `src/pages/Farmer/`: The Farmer AI Hub with crop calendar, irrigation watch, and agronomic chatbot.
- `src/context/`:
 - **LocationContext.tsx**: Manages active city coordinates (Delhi, Mumbai, Kanpur, Chennai, Ahmedabad).
 - **AuthContext.tsx**: Manages login sessions, Remember-Me, and user roles.
 - **FarmerContext.tsx**: Manages farmer profiles, crops, and land plots.
- `src/data/raiKnowledge/`: Contains **35,857 verified agronomic records** across 18 JSON files for the Farmer AI chatbot.

CHAPTER 5: FRONTEND WALKTHROUGH (WHAT THE USER SEES)

R.A.I. is organized into **5 Core Pillars**:

1. **Pillar 01 - Dashboard** (`/dashboard`):
 - Shows the Central Location Orb with live temperature, humidity, and weather conditions for the user's selected city.
2. **Pillar 02 - Interactive Risk Map** (`/risk-map`) [MAIN DEMO PAGE]:
 - Shows a live interactive map with **RainViewer Doppler radar precipitation overlays**.
 - Displays the **4-Layer Telemetry Inspector**:
 - **Layer 1: Observed** (Raw physical sensors: Temperature, Humidity, Pressure, Wind).
 - **Layer 2: Predicted** (Calibrated Heavy Rain Probability p%, Operational Threshold $\tau=0.015$).
 - **Layer 3: Explained** (TreeSHAP positive risk drivers in red, protective factors in blue).
 - **Layer 4: Operational Warning** (Official IMD-aligned action checklists).
3. **Pillar 03 - Emergency Civil Defense** (`/emergency`):
 - Displays river flood gauges, danger alert levels, and direct emergency helpline contacts (NDRF, State Disaster Authorities).
4. **Pillar 04 - Relief Logistics** (`/relief`):
 - Displays verified community relief centers, dry shelters, water supply hubs, and NGO coordination routes.
5. **Pillar 05 - Farmer Command Center** (`/farmer`):
 - Provides crop-stage-specific guidance (e.g. Sowing, Vegetative, Flowering, Harvesting) and irrigation advice grounded in soil type (e.g. Clayey vs. Sandy loam).

CHAPTER 6: BACKEND WALKTHROUGH (FASTAPI MADE SIMPLE)

Our backend runs on Python FastAPI at <http://127.0.0.1:8000>.

Key API Endpoints:

1. `GET /api/prediction/heavy-rainfall?latitude=...&longitude=...&city=...`
 - **What it does:** Ingests weather, runs XGBoost, computes TreeSHAP, and returns probability, risk level, and explanations.
2. `GET /api/model/status`
 - **What it does:** Returns model version (`v1.0.0-sih-xgb`), threshold (`0.015`), and training lineage.
3. `GET /api/model/explainability`
 - **What it does:** Returns the global ranking of all 27 atmospheric features based on mean absolute SHAP value.
4. `POST /api/auth/register/initiate`
 - **What it does:** Registers a new user, hashes password with salt + pepper, and sends a 6-digit OTP via Brevo email.
5. `POST /api/auth/otp/verify`
 - **What it does:** Verifies the OTP, marks the user as verified, and issues a 256-bit session token.
6. `POST /api/auth/login`
 - **What it does:** Authenticates returning users with brute-force rate limiting (5 attempts = 15-min lockout).

CHAPTER 7: LOGIN, OTP & SECURITY (HOW WE PROTECT USERS)

How OTP Verification Works:

1. User enters their email and password.
2. The server creates a secure random 6-digit code (e.g., `482910`).
3. The server computes a **Salted HMAC-SHA256 hash** of the code and saves it in the database with a 5-minute expiry timer.

- 4. The server calls the **Brevo Email API** using verified Sender ID **1** (RAINAIWORK@GMAIL.COM) to deliver the code to the user's inbox.
- 5. When the user enters the code, the server hashes the input and compares it in constant time (`timingSafeEqual`).
- 6. Once verified, the challenge is immediately destroyed (`is_consumed = TRUE`) so it can **never be reused** (Anti-Replay defense).

Remember Me & Password Safety:

- **Remember Me Checked:** Stores session in ``localStorage`` and saves only the non-sensitive email string in ``rai_remembered_email``.
- **Remember Me Unchecked:** Stores session in ``sessionStorage`` only (destroyed the second the tab is closed).
- **Passwords:** Are **NEVER stored** in ``localStorage``, ``sessionStorage``, cookies, or component state after submission.

CHAPTER 8: MACHINE LEARNING MADE SIMPLE (XGBOOST EXPLAINED)

1. What is the ML Goal?

Predict if a location will receive **Heavy Rainfall (>= 64.5 mm in 24 hours)**.
64.5 mm/day is the official threshold set by the India Meteorological Department (IMD).

2. Why XGBoost?

- **Analogy:** Imagine asking 100 experienced meteorologists to make a forecast. Each expert learns from the mistakes of the previous expert. That is what Gradient Boosted Decision Trees (XGBoost) does!
- It works exceptionally well on tabular atmospheric numbers (temperature, pressure, dew point) and runs super fast on standard CPUs without needing expensive GPU servers.

3. The 14 Key Weather Features:

Our model inspects 14 atmospheric indicators:

- 1. ``precipitation_sum``: Rain accumulated over the past 24 hours.
- 2. ``dew_point_2m_mean``: Measure of moisture in the lower air.
- 3. `relative_humidity_2m_mean`: How saturated the air is (close to 100% = rain imminent).
- 4. `surface_pressure_mean`: Air pressure (a sudden drop means a storm or depression is arriving).
- 5. `cloud_cover_mean`: Total cloud density.
- 6. `wind_speed_10m_max`: Peak wind gusts driving moisture convergence.
- 7. `apparent_temperature_mean`: Heat index (thermal energy in the air).
- 8. `precipitation_probability_max`: Peak rain likelihood from numerical ensembles.
- 9. `precipitation_probability_mean`: Average rain likelihood.
- 10. `precipitation_hours`: How many hours it has already been raining.
- 11. `temperature_2m_max`: Peak daily temperature.
- 12. `temperature_2m_mean`: Average air temperature.
- 13. `temperature_2m_min`: Minimum overnight temperature.
- 14. `et0_fao_evapotranspiration`: Evapotranspiration rate.

CHAPTER 9: MODEL NUMBERS & METRICS (WHAT THEY MEAN)

These are the exact verified numbers from our held-out test dataset:

Metric Name	Our Score	Simple English Meaning
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ROC-AUC	**0.9605**	**Overall Discrimination** : The model distinguishes between heavy rain days and normal days with 96% separation accuracy.
PR-AUC	**0.1082**	**Precision-Recall Area** : Because heavy rain happens only 0.44% of the time, our score of 0.1082 is a **24.6x improvement** over random guessing!
Calibrated Brier Score	**0.0044**	**Probability Accuracy** : Measures error between predicted probability and actual reality (Lower is better; 0.0044 is rated **EXCELLENT**).
Validation Recall (tau=0.015)	**52.1%**	**Flood Catch Rate** : The model successfully catches more than half of all extreme heavy rainfall events 24 hours in advance.
Validation Precision (tau=0.015)	**10.96%**	**Alert Reliability** : 1 out of every 9 alerts results in extreme heavy rain (≥ 64.5 mm), while the rest bring moderate-to-heavy showers.
F₂ Score	**0.2975**	Weights catching floods (Recall) twice as heavily as false alarms (Precision).

CHAPTER 10: WHY 1.5% PROBABILITY TRIGGERS AN ALERT (THRESHOLD TUNING)

[TIP] The Class Imbalance Secret (Crucial for Viva!)

- **Question**: "Why does a 1.5% probability trigger a High Risk alert? Isn't 1.5% very small?"
- **Simple Answer**:
- In normal weather data across India, extreme heavy rain (≥ 64.5 mm) is very rare - it happens on only **0.44% of days (1 in 227 days)**.
- If you use the standard default threshold of **50% (0.50)**, the model will say "NO RAIN" 365 days a year. It would have 99.5% accuracy, but **0% recall** (it would miss every single flood!).
- A calibrated probability of **1.5% (0.015)** means the risk is **3.4x higher than normal**.
- Therefore, we tuned our operational threshold to **tau = 0.015**, giving us a **52.1% flood catch rate** while triggering alerts on only 7% of days!

The 4 Risk Tiers:

- **LOW** ($p < 0.75\%$): Normal dry / light rain conditions.
- **MODERATE** ($0.75\% \leq p < 1.5\%$): Elevated moisture; heightened monitoring advised.
- **HIGH** ($1.5\% \leq p < 5.0\%$): **Threshold Breached [WATCH]** - High probability of heavy rain.
- **CRITICAL** ($p \geq 5.0\%$): **Severe Danger [WARNING]** - Potential cloudburst / flood.

CHAPTER 11: EXPLAINABLE AI & TreeSHAP

1. What is TreeSHAP?

- **Analogy**: Imagine a cricket team scores 300 runs. TreeSHAP is like calculating exactly how many runs were contributed by the opening batsman (+80), the captain (+110), and how many runs were lost due to slow batting (-20).
- For every weather prediction, TreeSHAP calculates the exact positive or negative contribution of each weather variable.

2. The Formula in Plain English:

Final Prediction = Base Average + Sum of all Feature Contributions

- **Base Value (ϕ_i):** -4.85 log-odds (the average baseline probability across India).
- **Positive Factors ($\phi_i > 0$):** Push risk higher (e.g. High Dew Point +0.41).
- **Negative Factors ($\phi_i < 0$):** Push risk lower (e.g. High Barometric Pressure -0.28).

3. Human-Readable Output:

R.A.I. translates math into simple English on the screen:

- **"High humidity (92%) and past rainfall (35mm) are driving flood risk up."**
- **"Moderate wind speed is helping disperse clouds, preventing sudden cloudburst."**

CHAPTER 12: WARNING SYSTEM & FARMER AI HUB

1. Operational Warning Levels:

- **`NO\_WARNING`:** Normal daily routines.
- **`ADVISORY`:** Check stormwater drains and clear debris.
- **`WATCH`:** High heavy rain risk. Farmers should pause fertilizer spraying; citizens avoid underpasses.
- **`WARNING`:** Severe danger. Emergency shelters activated; relief boats on standby.

2. Farmer AI Hub:

- Contains a knowledge base of **35,857 verified agronomic records** across 18 categories.
- Uses **1.75x Farmer Persona Boosting** so agricultural terms are prioritized for farmers.
- Understands **Hindi and Hinglish terms** (e.g. **sinchai** \to irrigation, **jalbharav** \to waterlogging).
- Grounds advice in the farmer's specific **soil type** (e.g. Black Cotton soil retains water, Sandy soil drains quickly) and **crop stage** (e.g. Flowering vs. Harvest).

CHAPTER 13: SATELLITE & WEATHER INGESTION

1. NASA GPM IMERG Satellite Grid

- **What it is:** NASA's constellation of rainfall satellites orbiting Earth.
- **Resolution:** $0.1^\circ \times 0.1^\circ$ (about 10 km x 10 km).
- **Role in R.A.I.:** Provides ground-truth satellite rainfall maps and antecedence moisture levels.

2. Open-Meteo Weather Model

- **What it is:** High-resolution numerical weather prediction model.
- **Role in R.A.I.:** Provides hourly forecasts for temperature, surface pressure, dew point, relative humidity, and wind.
- **Offline Resilience:** If the internet or external API is slow, R.A.I. automatically uses local seasonal climatology fallbacks so the app never crashes!

CHAPTER 14: SUPABASE DATABASE

Our cloud database runs on **Supabase PostgreSQL** with 6 relational tables:

1. **`users`:** Stores user ID, email, salted password hash, and account role (*`user`* vs *`farmer`*).
2. **`sessions`:** Stores 256-bit cryptographic login tokens and expiration dates.
3. **[otp_challenges](#):** Stores 6-digit OTP hashes, attempt counters (max 5), and 5-min expiry timers.
4. **[farmer_profiles](#):** Stores village, district, soil type, and irrigation source.
5. **[farm_plots](#):** Stores field names, acreage, active crop (e.g. Wheat, Rice), and coordinates.
6. **[user_preferences](#):** Stores preferred language (Hindi/English) and default city node.

CHAPTER 15: DOCKER & RUNNING THE APP LOCALLY

1. What is Docker?

Docker packages our entire backend (Python, FastAPI, XGBoost, models, and libraries) into a single container image so it runs identically on any computer without dependency errors.

2. How to Run R.A.I. Locally:

Terminal 1 - Start FastAPI Backend:

```
.\.venv\Scripts\Activate.ps1  
python -m uvicorn ml.server:app --host 127.0.0.1 --port 8000
```

Terminal 2 - Start React Frontend:

```
npm run dev
```

Open your browser at: <http://localhost:3000>

CHAPTER 16: "WHY DID WE USE THIS?" (JUDGES' FAVORITE QUESTIONS)

- Q: Why XGBoost instead of Deep Learning (CNN/LSTM)?

Answer: Weather data is tabular (numbers in tables). XGBoost consistently outperforms deep neural networks on tabular data, runs fast on CPUs without GPUs, and supports exact TreeSHAP math.

- Q: Why TreeSHAP instead of LIME?

Answer: LIME uses random approximations which can give different answers each time. TreeSHAP gives exact, deterministic mathematical values based on game theory.

- Q: Why do you calibrate probabilities?

Answer: XGBoost outputs distorted raw scores on imbalanced data. Calibration maps these numbers so that a predicted 10% probability actually means heavy rain occurs 10% of the time in the real world.

- Q: Why not use the standard 0.50 threshold?

Answer: Heavy rainfall occurs on only 0.44% of days. At 0.50, the model would predict zero rain every day and miss every flood. Setting $\tau = 0.015$ catches 52.1% of floods.

- Q: Why FastAPI instead of Django or Flask?

Answer: FastAPI is asynchronous, supports Pydantic automatic data validation, and generates interactive OpenAPI documentation out of the box.

CHAPTER 17: 100+ RAPID-FIRE VIVA QUESTIONS & 1-LINE ANSWERS

General & Project

1. **What is R.A.I.?** -> Rainfall Artificial Intelligence for SIH1521.
2. **What is the goal?** -> Predict heavy rainfall 24h ahead and explain why using XAI.
3. **What is the IMD heavy rain cutoff?** -> ≥ 64.5 mm in 24 hours.
4. **Who are the 3 users?** -> Citizens, Disaster Officials (NDRF), and Farmers.
5. **What are the 5 pillars?** -> Intelligence, Risk Map, Emergency, Relief, Farmer Hub.

Machine Learning & Data

6. **What model is used?** -> XGBoost with Isotonic / Platt probability calibration.
7. **How many training samples?** -> 175,440 hourly records across 10 Indian climate zones.
8. **How many features?** -> 14 core features in the production signature.
9. **What is the ROC-AUC?** -> 0.9605 (Exceptional global discrimination).
10. **What is the PR-AUC?** -> 0.1082 (24.6x lift over baseline).

11. **What is the Brier score?** -> 0.0044 (Rated EXCELLENT calibration).
12. **What is the operational threshold (tau)?** -> tau = 0.0150 (1.5% probability).
13. **What is the recall at tau=0.015?** -> 52.1% of heavy rainfall events caught.
14. **How was data split?** -> Chronologically (Train: 2023-May 2024, Val: May-Sep 2024, Test: Sep-Dec 2024).

Explainable AI (XAI)

15. **What XAI method is used?** -> TreeSHAP (SHapley Additive exPlanations).
16. **What is the base value phi_0?** -> -4.85 log-odds (average training baseline).
17. **What does a positive SHAP value mean?** -> The feature increased heavy rain risk.
18. **What does a negative SHAP value mean?** -> The feature reduced heavy rain risk.

Backend, Database & Security

19. **What web framework is used?** -> Python FastAPI on port 8000.
20. **What database is used?** -> Supabase Cloud PostgreSQL with 6 relational tables.
21. **How is OTP sent?** -> Via Brevo Transactional Email REST API (Sender ID 1).
22. **How are passwords stored?** -> Salted SHA-256 with server-side pepper.
23. **What happens after 5 failed logins?** -> 15-minute account lockout (Brute-force defense).
24. **How many automated tests passed?** -> 160 out of 160 tests (100% pass rate).

CHAPTER 18: TOP 20 MUST-KNOW QUESTIONS FOR THE PRESENTATION

1. What is your project problem statement?

Answer: SIH1521 - Explainable AI for Heavy/High-Impact Rainfall Prediction.

2. What counts as heavy rainfall?

Answer: ≥ 64.5 mm/day as defined by the India Meteorological Department (IMD).

3. What algorithm powers the prediction?

Answer: XGBoost Decision Trees with Platt/Isotonic Probability Calibration.

4. Why is your decision threshold 1.5% (tau = 0.015)?

Answer: Because heavy rain happens on only 0.44% of days; a 1.5% threshold catches 52.1% of floods while avoiding false alarm fatigue.

5. What is your model's ROC-AUC and PR-AUC?

Answer: ROC-AUC is 0.9605; PR-AUC is 0.1082 (24.6x lift over baseline).

6. How does your Explainable AI work?

Answer: TreeSHAP calculates exact additive feature attributions showing positive risk drivers and protective factors.

7. What are the 4 layers in the Telemetry Inspector?

Answer: Observed Weather $\rightarrow$ Predicted Probability $\rightarrow$ TreeSHAP Explanations $\rightarrow$ Operational Warnings.

8. Where does your weather data come from?

Answer: Open-Meteo High-Resolution NWP forecasts and NASA GPM IMERG 0.1 deg satellite grids.

9. How did you prevent data leakage during training?

Answer: Strict chronological train/validation/test splitting over 175,000 real hourly records.

10. What is the Brier score of your model?

Answer: 0.0044, demonstrating excellent probability calibration.

11. What backend framework is used?

Answer: Python FastAPI running on Uvicorn ASGI on port 8000.

12. What frontend technology is used?

Answer: React 18, TypeScript, Vite, and scoped CSS Modules on port 3000.

13. What database stores user data?

Answer: Supabase Cloud PostgreSQL with 6 relational tables and IDOR tenant isolation.

14. How does your OTP system work?

Answer: 6-digit cryptographic code, salted HMAC hash, single-use destruction, dispatched via Brevo email.

15. What is Farmer AI?

Answer: An agronomic hub with 35,857 knowledge records, 1.75x persona boosting, and Hindi/Hinglish synonym expansion.

16. How is password security enforced?

Answer: Salted SHA-256 with server pepper; zero passwords stored in browser storage.

17. How is the backend containerized?

Answer: Dockerfile using `python:3.11-slim` running as non-root user `raius:1000`.

18. How many automated tests did you run?

Answer: 160 automated tests across Pytest, live cloud E2E, auth hardening, and security audits (100% pass).

19. What happens if an external API goes down?

Answer: Automatic fallback to local seasonal climatology heuristics so the system never crashes.

20. What makes R.A.I. unique compared to standard weather apps?

Answer: It transforms black-box probabilities into transparent, physics-grounded civil actions that users can trust.

CHAPTER 19: 6-MEMBER TEAM ROLE DISTRIBUTION

Member 1: Team Lead & Presentation -> SIH1521 Mission, 5 Pillars, Live UI Walkthrough
 Member 2: ML & Data Lead -> XGBoost, Calibration, Training Splits, Metrics
 Member 3: Explainable AI Lead -> TreeSHAP Mathematics, Feature Attributions
 Member 4: Frontend Lead -> React 18, 4-Layer Inspector, RainViewer Radar
 Member 5: Backend & API Lead -> FastAPI, Pydantic Models, Open-Meteo Ingestion
 Member 6: Security & Cloud Lead -> Supabase PostgreSQL, Brevo OTP, Docker Container

CHAPTER 20: 1-PAGE MASTER CHEAT SHEET (MEMORIZE THIS!)

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| R.A.I. 1-PAGE VIVA MASTER CHEAT SHEET |
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| Problem Statement | SIH1521 - Explainable AI for Heavy Rainfall Prediction | |
| Heavy Rain Cutoff | >= 64.5 mm / 24 hours (IMD Standard) |
| Historical Freq. | 0.44% of days (Severe Class Imbalance: 1 in 227) |
| ML Algorithm | XGBoost + Platt / Isotonic Probability Calibration |
| Operating Point | tau = 0.015 (1.5% probability) -> 52.1% Flood Recall |
| ROC-AUC / PR-AUC | ROC-AUC: 0.9605 | PR-AUC: 0.1082 (24.6x Baseline Lift) |
| Brier Score | 0.0044 (Rated EXCELLENT Probability Reliability) |
| XAI Engine | TreeSHAP (Additive Shapley Feature Attributions) |
| 4-Layer Inspector | Observed -> Predicted -> Explained -> Operational Action |
| Data Feeds | Open-Meteo NWP + NASA GPM IMERG 0.1 deg Satellite Grid |
| Full Stack Ports | React 18 (Port 3000) <-> Python FastAPI (Port 8000) |
| Cloud Database | Supabase Managed PostgreSQL (6 Relational Tables) |
| OTP Delivery | Brevo Transactional Email REST API (Sender ID: 1) |
| Farmer AI Hub | 35,857 Records, 1.75x Persona Boost, Hinglish Synonyms |
| Containerization | Docker (python:3.11-slim, Non-Root rauser:1000) |
| Verified Tests | 160 / 160 Automated Tests Passed (100% Pass Rate) |
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